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Adaptive biomimicry: design of neural interfaces with enhanced biointegration

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Neural interfaces (NIs) have traditionally used inorganic device constructs paired with electrical stimulation to bypass injured or diseased electroactive tissues. These bioinert devices have significant impact on the neural tissue, being synthetic and causing large volumetric changes to the biological environment. The concept of biomimicry has become popular for tissue engineering technologies, reflecting biological properties as a component of material design. Tissue engineering strategies can be harnessed in bioelectronic device design to improve biological tolerance, but the need for improved integration with the native tissue remains an unmet need. Adaptive biomimetic designs that respond to the changing neural tissue environment associated with wound healing can actively address the immune response to improve biointegration. These adaptive approaches include responsive materials paired with stem cells and bioactive molecules as integrated components of NIs. Combining adaptive biomimetics with NIs provides a new, more natural approach for communicating with the nervous system.

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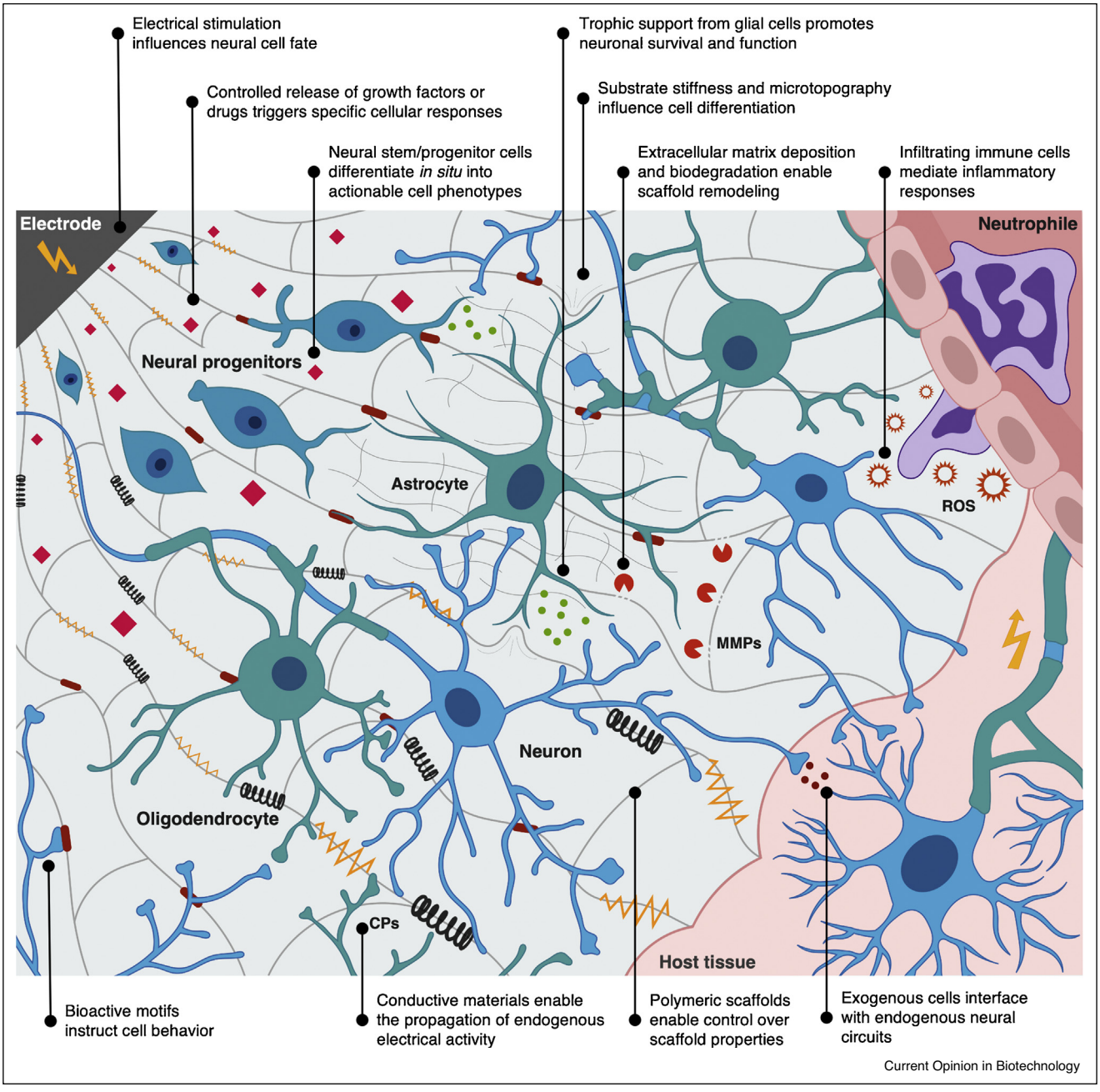
Biomimetics refers to the rational emulation of natural systems and constitutes the basis for biologically inspired engineering. In the biomedical context, one of the main applications of biomimetic design has been the development of artificial constructs that mimic physiological tissues [1•,2]. Biomimetic tissues have been extensively used in basic and translational research, with applications ranging from drug screening [3] and disease modelling [4] to biomedical devices [5••,6] and regenerative medicine [7]. However, physiological tissues are characterized by a uniquely dynamic state of equilibrium with high spatio-temporal variability. This tissue homeostasis is maintained *in vivo* by a complex interplay of endogenous and exogenous factors that drive cell survival, proliferation and differentiation [8••,9]. Furthermore, individual cells are surrounded by an intricate milieu of physicochemical cues that fluctuate in response to different stimuli [2]. Although this state of constant adaptation underlies normal tissue development, repair and regeneration, it also presents several challenges for biomimetic tissue engineering (TE).

Conventional biomimetic technologies have been developed through bottom-up design, which refers to the isolation and study of individual bioinspired features to be applied to synthetic systems [10,11]. However, this fixed approach does not account for the complexity and dynamism that underlie normal tissue physiology. Alternatively, top-down design strategies that tap into and exploit endogenous processes involved in tissue homeostasis are increasingly being explored. This adaptive approach to biomimicry could bridge the divide between synthetic and biological systems, which often impairs the biointegration of implantable TE technologies. This issue is particularly critical for the efficacy and long-term stability of devices that directly interface with the nervous system [9,12,13]. Nervous tissues are remarkably sensitive to homeostatic perturbations elicited by injury or trauma, owing to their soft, electroactive and highly organized nature. In addition, implantation invariably triggers physiological mechanisms involved in inflammation and wound healing, which are orchestrated by the host immune system to restore tissue homeostasis [14]. These local oscillations in physiological parameters generate a distinct post-injury microenvironment (PIM), which greatly influences cellular and molecular phenomena at the device-tissue interface (Figure 1). For instance, infiltrating neutrophils produce oxidative and proteolytic enzymes that prime the wound site for repair [15•,16]. However, sustained neutrophile recruitment can also exacerbate tissue trauma and trigger inflammatory responses. Biotechnological strategies based on the modulation of endogenous immune responses have been increasingly explored and demonstrated to improve cancer treatment [17–19] and immunological tolerance of transplants [20,21]. This evidence suggests that dynamic and more stable device-tissue interactions could be established by harnessing the inherent complexity and amplification present in biological systems. Therefore, adaptive biomimetic strategies hold great potential to engineer

Figure 1

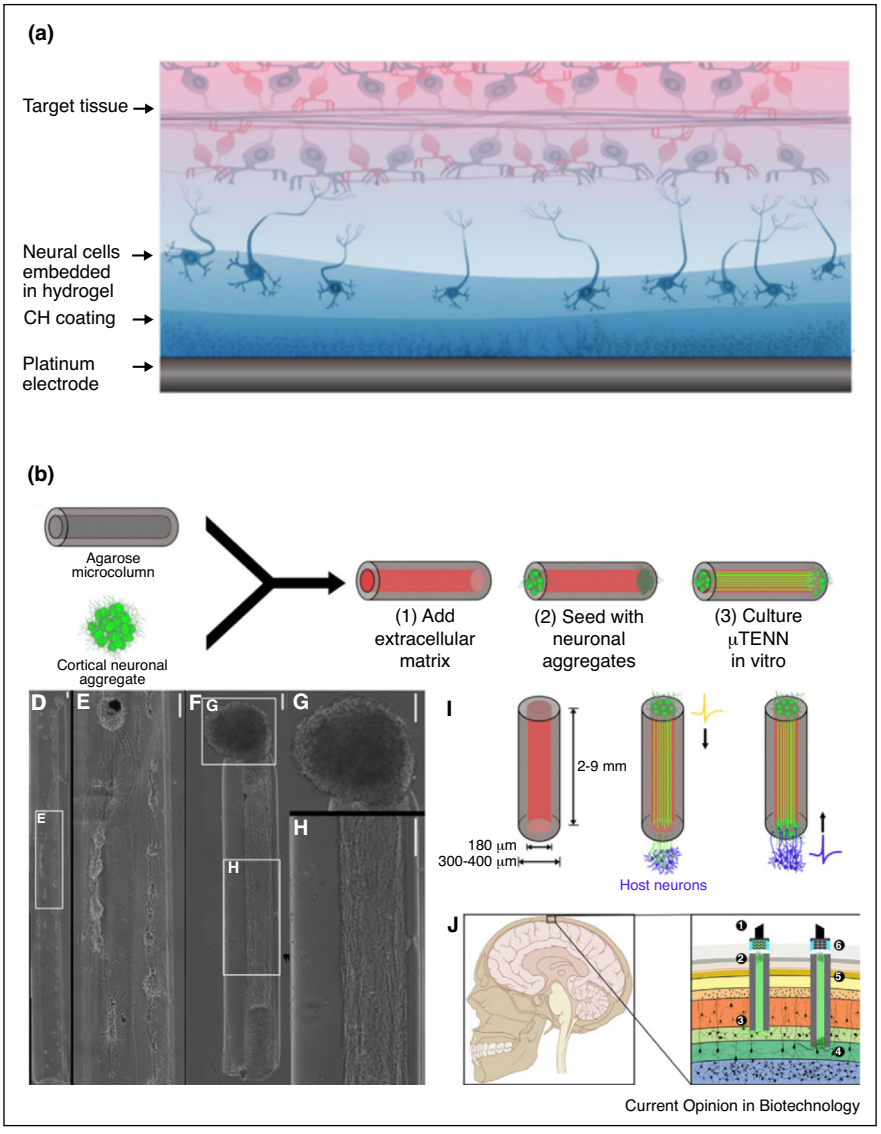


Tissue homeostasis and biointegration of neural interfaces.
The schematic depicts the complex interplay between the cellular, biochemical, and biophysical components of the post-injury microenvironment, which mediate the biointegration of implantable neural interfaces. (ROS: Reactive oxygen species, MMPs: Matrix metalloproteinases, CPs: Conductive polymers).

local microenvironments that foster the biointegration of implantable neurotechnologies.

Neural interfaces (NIs) that enable bidirectional communication between the nervous system and external devices have gathered both research and commercial interest [22]. These technologies are being increasingly implemented in the clinic and proposed for a variety of emergent non-clinical applications. NIs rely on the use of electrodes to interface with nervous tissues and record or stimulate the activity of discrete populations of neurons [23]. Conventional strategies based on rigid and inorganic

Figure 2



(a) Multi-layered living electrode coating applied to a Neuronexus MEA. Reproduced with permission [6]. **(b)** Fabrication and implantation of microtissue engineered neural networks (μTENN). Reproduced with permission [5••].

materials have been largely associated with poor biointegration and foreign body responses (FBRs) that negatively impact device performance [24•,25]. Biomimetic strategies that reproduce key features of physiological tissues have been proposed to enhance device biocompatibility and performance. One salient example is the development of tissue engineered 'living electrodes' (LEs), which are based on the integration of bioactive molecules, neural cells and biomaterials with defined physicochemical properties [26]. These technologies are aimed at establishing functional connections with the host nervous system to harness endogenous mechanisms of synaptic modulation. The development of LEs could address the major challenges associated with chronically stable NIs by minimising FBRs and enabling sustained long-term performance. Furthermore, the ability to deliver combinations of cells and physicochemical cues in response to time-dependent microenvironmental changes holds great potential to engineer homeostatic states that promote biointegration.

One embodiment of LEs is based on the integration of a non-degradable conductive hydrogel (CH) and a degradable cell-laden biosynthetic hydrogel [6]. This layered construct can be applied to conventional neural microelectrode arrays (MEAs) as a coating (Figure 2a). A CH

comprised of modified polyvinyl alcohol and poly(3,4-ethylenedioxythiophene) (PEDOT) improves the charge transfer capacity of the underlying platinum electrodes, while also reducing the strain mismatch between the MEA and the nervous tissue. The biosynthetic degradable hydrogel supports the formation of functional neural networks, which can synapse with the native tissues to drive the NI [27••,28•]. To achieve this, multiple biomimetic cues are incorporated within the biosynthetic construct to guide cell fate, including stiffness, topography, growth factors, neuroprotective factors and cell adhesion motifs [14,29]. The use of a biodegradable component allows for programable biomimetic stimuli that can be tuned to match the dynamic processes that occur upon implantation. Another example of LEs developed by Adewole *et al.* [5••] relies on microtissue engineered neural networks (μTENN) to enable natural synaptic integration with target tissues (Figure 2b). μTENNs are self-contained implantable channels that consist of discrete populations of optogenitcally modified neurons, which are connected by long bundles of axons within a microscopic hydrogel column. The formation of synapses between the implanted axons and host neurons enables bidirectional communication across the NI upon implantation. The neuronal aggregates remain accessible at the surface of the brain, allowing μTENNs to act as biologically inspired relays that enable optical stimulation and visual recording of neuronal activity. The implementation of biomimetic TE technologies capable of creating functional synapses with native tissues has the potential to fundamentally alter the way in which neuroprosthetic devices communicate with the nervous system.

Physiological tissues such as the brain behave as nonlinearly elastic materials that exhibit characteristic viscoelasticity and plasticity [30]. Biomimetic hydrogel-based systems could also be tuned to adapt to the stiffness and relaxation behaviour of soft biological tissues. For instance, Tringides *et al.* [31] recently reported the development of electrode arrays based on an ultrasoft viscoelastic hydrogel matrix, which was loaded with conductive carbon nanomaterials. These viscoelastic arrays were shown to plastically deform upon implantation and readily conform to the underlying tissue, without compromising their electrical and mechanical performance. Furthermore, a maximum signal to noise ratio of 18 was obtained, which enabled the recording of low-amplitude local field potentials. Soft and flexible NIs also hold great potential to reduce tissue damage associated with surgical delivery, as well as throughout chronic implantation periods. For instance, Park *et al.* [32] developed fully polymeric neural probes with low stiffness to reduce tissue damage associated with the relative motion of the brain with respect to the skull. Their results showed that polymer fiber probes elicited significantly lower tissue responses and blood-brain barrier breach after three days of implantation, when compared to similarly sized insulated steel microwires. Furthermore, immunohistochemical evaluation demonstrated that the polymeric probes led to reduced expression of biomarkers associated with glial scarring for up to three months following implantation. Taken together, these results highlight the potential benefits of incorporating biomimetic design criteria in the development of chronically stable NIs with sustained performance.

Exogenous cells do not only act as relays for synaptic communication, but they can also provide biochemical stimuli that can mediate the formation of chronically stable NIs. Cell replacement therapies based on stem and progenitor cells enable the supply of neuroprotective, neuroregenerative and immunomodulatory factors [33,34]. Neural differentiation and electrophysiological maturation are largely influenced by neurotrophic factors produced by neurons and accessory cells, such as astrocytes [28•,35•]. As such, cell-based approaches constitute an ideal route for the delivery of therapeutics including nucleic acids for *in situ* protein expression, as well as genome and metabolic engineering [36•,37,38]. While optogenetics constitute the most common form of genetic manipulation applied to NIs, alternative genetic engineering approaches have been recently explored. Pinyon *et al.* proposed the use of neurotrophin gene augmentation via electrotransfer of plasmid DNA to improve the performance of cochlear implants [39,40•]. Bionic array Directed Gene Electrotransfer (BaDGE®) involves the use of modified cochlear electrode arrays to deliver plasmid DNA for *in situ* expression of brain derived neurotrophic factor (BDNF). Cochlear expression of BDNF promotes regrowth of spiral ganglion peripheral neurites towards the implanted electrode array, which lowers stimulation thresholds and improves the dynamic range of the cochlear nerve. Current cochlear implant devices are unable to deliver the required electrical stimulus for electroporation without exceeding the charge injection limit of the platinum electrodes [39]. However, the incorporation of neurotrophin gene augmentation into the implantation strategy is expected to improve biointegration by promoting regenerative neural growth and ultimately, improving sound perception for cochlear implant recipients.

Implantable NIs are also well poised to take advantage of stimulation strategies to improve biointegration, since they rely on electrodes to achieve their primary therapeutic effect. Electrical stimulation (ES) has been shown to influence the fate and function of electroactive tissues *in vitro* and *in vivo* [41••,42,43]. Endogenous electrical fields provide crucial morphogenetic cues during neural development and wound healing [44]. In addition, exogenous electrical fields guide neuronal migration and growth, while also influencing cytokine release, surface ligand presentation and extracellular matrix production by glial cells [43]. In particular, ES has been shown to

induce axon sprouting through the upregulation of mTOR signalling, while also promoting synapse formation via the Jak/Stat pathway [45]. ES has also been used to mobilize and modulate the biology of endogenous neural stem/progenitor cells (NSPCs) [46]. Furthermore, the delivery of biophysical stimuli such as ES does not contribute to exacerbate the FBR elicited by implantable neural devices, as opposed to insertion trauma and physicochemical mismatches [47]. This evidence suggests that ES-mediated remodelling of endogenous neural networks holds great potential to enhance the biointegration of NIs. Furthermore, the combination of ES with neuroregenerative and neuroprotective strategies could enable precise control over endogenous neuroplastic mechanisms for different therapeutic applications.

The implantation of NIs triggers tissue responses that are mediated by highly specialized multicellular phenomena [48]. These innate mechanisms drive the formation of glial borders to isolate foreign entities that disturb tissue integrity. The responses triggered by highly biocompatible materials appear morphologically similar to normal glia limitans, such as the meninges in the CNS [49••]. However, local inflammatory responses are often triggered by infiltrating monocytes and other immune cells upon breach of the blood brain barrier [9]. Biointegration of NIs could be enhanced by biomimetic strategies that harness these endogenous tissue responses. Although inflammation is traditionally perceived as detrimental for the success of implantable devices, inflammatory cells also modulate neuroprotective mechanisms that promote neural survival, growth and plasticity [50]. Furthermore, nervous tissues establish a high degree of crosstalk with the immune system owing to the significant overlap between soluble signals and molecular pathways [51••]. Because of this, the delivery of anti-inflammatory drugs could inadvertently impair the function of endogenous cells that mediate tissue repair and increase the risk of infection. This evidence highlights the need for mechanistic studies to identify biochemical and biophysical factors that modulate the biology of neural and immune cells. Moreover, bioinspired mechanisms of injury and disease could be harnessed to modulate immune responses that promote biointegration.

Emerging neurotechnologies represent a growing change in theoretical and experimental strategies towards engineering biomimetic NIs. From a device perspective, this reflects a shift from traditional inert materials towards more versatile alternatives. Biomimetic systems that are more dynamic and adaptable could facilitate translation into clinically relevant and commercially viable technologies. However, this will also present new challenges from a regulatory perspective owing to the incorporation of multiple biological, biochemical and biophysical agents into a single therapeutic strategy [52••]. This highlights the need for experimental design strategies that account for the complexity of the *in vivo* setting, while simultaneously minimising risk to bridge the translational gap. For instance, ES-based approaches could be used to circumvent the need for exogenous cell delivery by recruiting and modulating the biology of endogenous NSPCs [46,53•]. In addition, ES holds great potential to regulate the activity of glial cells involved in device encapsulation following FBR [46]. Bioinstructive materials with defined degradation kinetics and micro/nanotopologies could be used to guide neural fate and differentiation without the need for growth factors or drugs [1•,2,54]. In addition, engineering biochemical microenvironments via transgene delivery could guide the formation and biointegration of functional neural networks *in situ* [55•,56].

Future NI research should focus on gaining mechanistic insight of tissue-device interactions to identify actionable pathways that foster biointegration. Signalling mechanisms for neuroimmune crosstalk are of particular interest, since embryonic stem cell proliferation and neurogenesis are modulated by the immune system [57,]. This approach could lead to the discovery of master regulators that moderate the communication between NSPCs and the immune system in order to establish biointegrative microenvironments. Although conventional *in vitro* and *in vivo* models will continue to be relevant for early screening and clinical validation, respectively, there is a need for experimental tools that enable the study of complex physiological phenomena. New methodologies are needed to visualize and assess synaptic connectivity within neural networks at the device-tissue interface [35•,58•]. In addition, technological developments are needed to deliver, operate and power the next generation of NIs, as well as connecting them to external devices and actuators. Therefore, conventional design strategies should be revisited to pre-emptively address emerging technical challenges in order to realise the full potential of future neurotechnologies.

Conflict of interest statement

Nothing declared.

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- • of special interest
- •• of outstanding interest

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